Student Portfolio

Learning Activities During Four Year UG Program

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1 Introduction

This Learning Portfolio is a reflective and evidence-based record of my academic journey, showcasing the skills, knowledge, and competencies I have developed through various curricular and co-curricular activities. It highlights not only the progress I have made in mastering subject-specific content but also my ability to apply this learning in practical, real-world contexts.

The activities and experiences documented in this portfolio are thoughtfully aligned with the United Nations Sustainable Development Goals (SDGs) and the Program Outcomes (POs) defined by the institution. By engaging in projects that promote sustainability, innovation, ethical practices, and lifelong learning, I have contributed meaningfully to goals such as SDG 3, SDG 4, SDG 7, SDG 8, SDG 9, SDG 10, SDG 11, SDG 12.

Simultaneously, the portfolio demonstrates growth in core Program Outcomes like engineering knowledge, problem-solving, ethical responsibility, and communication skills. These outcomes reflect the competencies expected of a globally responsible and professionally capable graduate.

This portfolio stands as a testament to my holistic development as a learner and a future professional, guided by both institutional excellence and global development priorities.

2 Learning Activities Overview

This section provides a consolidated overview of the key learning activities undertaken during the academic term. These activities span across academic projects, hands-on laboratory work, workshops, internships, community service, seminars, and co-curricular engagements. Each activity has been thoughtfully designed to enhance subject knowledge, practical skills, critical thinking, and social responsibility.

The learning activities are aligned with both the Sustainable Development Goals (SDGs) and the Program Outcomes (POs), ensuring that each experience contributes to academic excellence as well as global and societal development. The detailed list of activities undertaken during undergraduate program presented in Table 1.

Table 1: Detailed list of activities undertaken during undergraduate program.

#	**Activity**	**Title of the Activity**	**SDGs Addressed**	**POs/PSOs Mapping**
1	Curriculum Courses	List courses undertaken, their importance, and knowledge gained.	SDG 4, SDG 7	PO1, PO2
2	Case Studies	Brief description of the case studies explored and key learnings.	SDG 9, SDG 11	PO1, PO2, PO3
3	Workshops and Industrial Visits	Insights from hands-on experiences and industrial exposure.	SDG 9, SDG 16	PO1, PO2, PO3
4	Mini Project	Focus, challenges, and outcomes.	SDG 3	PO1, PO2, PO4, PO5

#	Activity	Title of the Activity	SDGs Addressed	POs/PSOs Mapping
5	Cultural and Sports Activities	Participation in cultural and sports events.	SDG 3, SDG 11	PO1, PO2, PO3
6	Internships and Trainings	Industry-based training and practical skills.	SDG 8, SDG 11	PO1, PO12
7	Value-Added Courses	Additional courses/certifications pursued beyond curriculum.	SDG 4, SDG 8, SDG 11	PO1, PO2
8	Course-End Projects	Projects carried out at the end of individual courses.	SDG 3, SDG 7, SDG 9 SDG 11	PO3, PO5, PO7
9	Social Service and Community Projects	Volunteering for social causes and community development.	SDG 2, SDG 12	PO3, PO5
10	Major Project	Description of final-year project and key contributions.	SDG 4, SDG 10	PO1, PO2, PO5, PO10
11	Research Work and Publications	Research papers and publications presented at conferences or in journals.	SDG 7, SDG 9, SDG 12	PO1, PO2, PO5, PO10

3 Mapping Activities to SDGs and Program Outcomes

3.1 Curriculum Courses

Course Title: Linear Algebra and Ordinary Differential Equations
SDG(s): SDG 4, SDG 9
Program Outcomes: PO1, PO2, PO5
Description: This course covered the fundamentals of linear algebra and ordinary differential equations, focusing on their applications in engineering problems such as system modeling and analysis.
Impact: Improved problem-solving abilities and a deeper understanding of mathematical tools used in engineering analysis.
Skills Developed: Analytical thinking, mathematical modeling, problem-solving.

Course Title: Semiconductor Physics
SDG(s): SDG 7, SDG 9, SDG 13
Program Outcomes: PO1, PO2, PO5
Description: This course explored the behavior of semiconductor materials, their electrical properties, and applications in modern electronics, focusing on the principles governing devices like transistors, diodes, and solar cells.
Impact: Enhanced understanding of semiconductor technologies, critical for developing

energy-efficient devices and advancing electronic innovations.
Skills Developed: Problem-solving, material analysis, understanding of semiconductor devices, and use of modern tools for electronic simulations.

Course Title: English for Communication
SDG(s): SDG 4
Program Outcomes: PO10, PO9
Description: This course focused on enhancing communication skills, including listening, speaking, reading, and writing, to effectively communicate in professional and academic environments.
Impact: Improved ability to communicate clearly and professionally in both written and verbal forms.
Skills Developed: Effective communication, presentation skills, writing proficiency, and listening comprehension.

Course Title: Problem Solving through C
SDG(s): SDG 4, SDG 9
Program Outcomes: PO1, PO2, PO5
Description: This course focused on developing problem-solving skills using the C programming language, covering fundamental programming concepts such as loops, arrays, functions, and algorithms.
Impact: Enhanced ability to approach and solve complex problems through systematic programming techniques.
Skills Developed: Programming skills, algorithm development, logical thinking, and debugging.

Course Title: Basic Electrical Engineering
SDG(s): SDG 7, SDG 9
Program Outcomes: PO1, PO2, PO5
Description: This course introduced the fundamentals of electrical engineering, covering topics such as electrical circuits, network analysis, electromagnetism, and power systems.
Impact: Gained foundational knowledge in electrical circuits and systems, critical for understanding more advanced engineering concepts.
Skills Developed: Circuit analysis, problem-solving, and understanding of electrical components and systems.

Course Title: Engineering Workshop and Computer Aided Drawing
SDG(s): SDG 4, SDG 9
Program Outcomes: PO3, PO5, PO9
Description: This course provided practical training in workshop techniques and computer-aided drawing (CAD) tools, focusing on the design, modeling, and visualization of engineering components.
Impact: Developed hands-on skills in workshop practices and CAD software, essential for designing and manufacturing engineering products.
Skills Developed: Workshop skills, CAD proficiency, design thinking, and visualization

of engineering components.

Course Title: Social Innovation
SDG(s): SDG 8, SDG 10
Program Outcomes: PO6, PO7, PO12
Description: This course focused on the application of innovative solutions to address social issues, with an emphasis on sustainable development, social entrepreneurship, and community-based projects.
Impact: Encouraged the development of socially responsible and sustainable solutions, fostering a mindset of innovation for societal impact.
Skills Developed: Problem-solving, social entrepreneurship, sustainable development, and community engagement.

Course Title: Advanced Calculus and Partial Differential Equations
SDG(s): SDG 4, SDG 9
Program Outcomes: PO1, PO2, PO5
Description: This course covered advanced topics in calculus and partial differential equations, focusing on their application to engineering and physical problems.
Impact: Gained in-depth knowledge of advanced mathematical techniques for solving complex engineering problems.
Skills Developed: Advanced problem-solving, mathematical modeling, and analytical skills in calculus and differential equations.

Course Title: Engineering Chemistry
SDG(s): SDG 4, SDG 9
Program Outcomes: PO1, PO2, PO5
Description: This course introduced the fundamental principles of chemistry, including topics like chemical bonding, thermodynamics, and materials science, with applications in engineering.
Impact: Developed a solid understanding of chemical processes and materials, crucial for engineering design and analysis.
Skills Developed: Analytical skills, understanding of chemical properties, and practical knowledge of materials and processes in engineering.

Course Title: Business English
SDG(s): SDG 4
Program Outcomes: PO10, PO9
Description: This course focused on developing communication skills in a business context, including professional writing, presentations, and cross-cultural communication.
Impact: Improved business communication, helping students to effectively engage in professional and corporate settings.
Skills Developed: Business writing, presentation skills, professional communication, and cross-cultural understanding.

Course Title: Data Structures
SDG(s): SDG 4, SDG 9
Program Outcomes: PO1, PO2, PO5
Description: This course introduced fundamental data structures such as arrays, linked lists, stacks, queues, trees, and graphs, along with algorithms for their manipulation and optimization.
Impact: Developed a strong foundation in organizing and managing data, which is essential for efficient algorithm design and problem-solving in software development.
Skills Developed: Problem-solving, algorithm design, understanding of data structures, and programming skills.

Course Title: Electronic Devices and Circuits
SDG(s): SDG 7, SDG 9
Program Outcomes: PO1, PO2, PO3, PO5
Description: This course focused on the working principles and applications of semiconductor devices such as diodes, transistors, and operational amplifiers, along with the analysis and design of electronic circuits.
Impact: Enhanced understanding of electronic devices and circuits, critical for designing and troubleshooting electronic systems in various applications.
Skills Developed: Circuit analysis, device-level understanding, and hands-on experience in designing and simulating electronic circuits.

Course Title: Digital Logic Design
SDG(s): SDG 4, SDG 9
Program Outcomes: PO1, PO2, PO5
Description: This course covered the fundamentals of digital systems, including number systems, Boolean algebra, logic gates, combinational and sequential circuits, and the design of digital systems.
Impact: Developed skills in designing and analyzing digital systems, which are fundamental to electronics, computing, and embedded systems.
Skills Developed: Logic design, Boolean simplification, circuit design, and problem-solving using digital components.

Course Title: Engineering Exploration
SDG(s): SDG 4, SDG 9
Program Outcomes: PO1, PO3, PO9
Description: This course provided an overview of various engineering disciplines, offering insights into the engineering design process, problem-solving techniques, and the interdisciplinary nature of engineering projects.
Impact: Helped in developing a broader understanding of engineering fields, preparing students for a multidisciplinary approach to engineering challenges.
Skills Developed: Critical thinking, problem-solving, teamwork, and exposure to various engineering technologies and methodologies.

Course Title: Special Functions and Complex Variables
SDG(s): SDG 4, SDG 9
Program Outcomes: PO1, PO2, PO5
Description: This course focused on the study of special mathematical functions such as Gamma functions, Bessel functions, and the theory of complex variables, which are essential for solving complex engineering problems.
Impact: Gained advanced mathematical skills for solving engineering problems, particularly in fields like electromagnetics, fluid dynamics, and control systems.
Skills Developed: Mathematical modeling, problem-solving with special functions, and advanced calculus skills in the context of engineering applications.

Course Title: Electronic Circuit Analysis
SDG(s): SDG 7, SDG 9
Program Outcomes: PO1, PO2, PO3
Description: This course focused on the analysis and design of electronic circuits, including methods for solving DC and AC circuits, network theorems, and transient analysis of circuits.
Impact: Enhanced understanding of circuit behavior, which is fundamental for designing reliable and efficient electronic systems.
Skills Developed: Circuit analysis, problem-solving with network theorems, and simulation of electronic circuits.

Course Title: Signals and Systems
SDG(s): SDG 4, SDG 9
Program Outcomes: PO1, PO2, PO5
Description: This course covered the fundamental concepts of signals and systems, including signal representation, system analysis, Fourier transforms, and Laplace transforms.
Impact: Provided a solid foundation for analyzing continuous and discrete signals, essential for communications, control systems, and signal processing.
Skills Developed: Signal processing, system analysis, Fourier and Laplace transform techniques, and problem-solving in the context of engineering systems.

Course Title: Probability Theory and Stochastic Processes
SDG(s): SDG 4, SDG 9
Program Outcomes: PO1, PO2, PO5
Description: This course introduced probability theory, random variables, and stochastic processes, with applications to signal processing, communication systems, and reliability analysis.
Impact: Developed a strong understanding of probability and random processes, crucial for analyzing systems with uncertainty and noise in engineering applications.
Skills Developed: Probability modeling, stochastic analysis, understanding random processes, and applications in communication and signal processing.

Course Title: Digital Design through Verilog HDL
SDG(s): SDG 4, SDG 9
Program Outcomes: PO1, PO2, PO5
Description: This course focused on the use of Verilog Hardware Description Language (HDL) for digital system design, covering topics such as combinational and sequential logic design, testing, and simulation.
Impact: Gained hands-on experience in designing and simulating digital circuits using Verilog HDL, essential for FPGA and ASIC design.
Skills Developed: Verilog programming, digital circuit design, simulation, and hardware verification techniques.

Course Title: Universal Human Values 2: Understanding Harmony
SDG(s): SDG 4, SDG 16
Program Outcomes: PO6, PO8
Description: This course focused on understanding the importance of human values, ethics, and harmony in life, fostering an awareness of social, environmental, and cultural harmony in professional and personal contexts.
Impact: Developed a strong ethical foundation, promoting responsible behavior and decision-making that respects societal and environmental values.
Skills Developed: Ethical reasoning, social responsibility, awareness of societal issues, and conflict resolution.

Course Title: Engineering Design Thinking
SDG(s): SDG 9, SDG 12
Program Outcomes: PO3, PO7, PO9
Description: This course introduced the principles of design thinking, emphasizing user-centered design, creativity, problem-solving, and iterative prototyping in engineering design.
Impact: Enhanced problem-solving skills by focusing on innovative and sustainable solutions that meet user needs and environmental concerns.
Skills Developed: Creative thinking, prototyping, teamwork, and problem-solving with a focus on sustainability and user-centric solutions.

Course Title: Quantitative Aptitude
SDG(s): SDG 4
Program Outcomes: PO1, PO2
Description: This course focused on enhancing mathematical problem-solving skills through topics such as number systems, algebra, arithmetic, and data interpretation.
Impact: Strengthened analytical and quantitative skills, essential for engineering problem-solving and competitive exams.
Skills Developed: Analytical thinking, problem-solving, and proficiency in quantitative reasoning and mathematical techniques.

Course Title: Managerial Economics and Financial Analysis
SDG(s): SDG 8, SDG 9
Program Outcomes: PO11, PO12
Description: This course covered the basics of economics, financial management, and decision-making techniques, including cost analysis, investment evaluation, and financial statements.
Impact: Provided a solid understanding of economic and financial principles, essential for effective decision-making and project management in engineering projects.
Skills Developed: Economic analysis, financial modeling, budgeting, and decision-making under uncertainty.

Course Title: Computer Organization and Operating Systems
SDG(s): SDG 4, SDG 9
Program Outcomes: PO1, PO2, PO5
Description: This course introduced the architecture of computer systems, including the organization of processors, memory, I/O systems, and the fundamentals of operating systems such as process management, scheduling, and file systems.
Impact: Developed a deep understanding of how computers operate at the hardware and software levels, which is essential for optimizing system performance and troubleshooting.
Skills Developed: Computer architecture, operating system concepts, memory management, and system-level programming.

Course Title: Pulse and Digital Circuits
SDG(s): SDG 4, SDG 9
Program Outcomes: PO1, PO2, PO3
Description: This course covered the analysis and design of pulse circuits, multivibrators, and digital circuits, focusing on applications such as signal generation, logic circuits, and timing circuits.
Impact: Gained a strong understanding of pulse circuits and their applications in digital systems, which are critical for communication, control, and signal processing.
Skills Developed: Pulse circuit design, digital logic, waveform analysis, and application of digital components in various systems.

Course Title: Analog Communications
SDG(s): SDG 9
Program Outcomes: PO1, PO2, PO5
Description: This course focused on the fundamentals of analog communication systems, covering topics such as amplitude modulation, frequency modulation, and signal transmission over communication channels.
Impact: Gained a strong understanding of analog communication techniques, which are essential for the design and analysis of wireless communication systems.
Skills Developed: Analog modulation techniques, signal analysis, and understanding of communication systems' performance.

Course Title: Linear Control Systems
SDG(s): SDG 9, SDG 12
Program Outcomes: PO1, PO2, PO3, PO7
Description: This course introduced the theory and analysis of linear control systems, covering topics such as transfer functions, stability analysis, feedback systems, and control design techniques.
Impact: Developed an understanding of control system design and analysis, crucial for applications in automation, robotics, and embedded systems.
Skills Developed: Control system modeling, stability analysis, system response analysis, and feedback control design.

Course Title: Electromagnetic Theory and Transmission Lines
SDG(s): SDG 7, SDG 9
Program Outcomes: PO1, PO2, PO5
Description: This course focused on the fundamental principles of electromagnetism, including Maxwell's equations, electromagnetic wave propagation, and the behavior of transmission lines.
Impact: Developed an in-depth understanding of electromagnetic fields and transmission line theory, essential for the design and analysis of communication systems and antennas.
Skills Developed: Electromagnetic theory, transmission line analysis, wave propagation, and application in communication systems and antenna design.

Course Title: Product Realization
SDG(s): SDG 9, SDG 12
Program Outcomes: PO3, PO9, PO11
Description: This course covered the processes involved in taking a product from concept to market, including design, prototyping, manufacturing, testing, and commercialization.
Impact: Provided insights into the complete product lifecycle, which is crucial for engineers involved in product development, ensuring efficiency and quality.
Skills Developed: Product design, prototyping, manufacturing process understanding, and commercialization strategies.

Course Title: Analytical Reasoning
SDG(s): SDG 4
Program Outcomes: PO2, PO12
Description: This course focused on enhancing logical and analytical thinking through exercises in reasoning, problem-solving, and decision-making across various contexts.
Impact: Strengthened the ability to analyze complex problems systematically, a skill crucial for both academic and professional success in engineering.
Skills Developed: Logical thinking, pattern recognition, problem-solving, and critical analysis.

Course Title: Environmental Science
SDG(s): SDG 13, SDG 15
Program Outcomes: PO7, PO6
Description: This course covered the fundamentals of environmental science, including ecosystems, biodiversity, pollution, and the impact of human activities on the environment.
Impact: Raised awareness of environmental issues and sustainability practices, providing a strong foundation for designing eco-friendly engineering solutions.
Skills Developed: Environmental analysis, sustainability practices, and understanding of ecological systems and their protection.

Course Title: Integrated Circuits and Applications
SDG(s): SDG 9
Program Outcomes: PO1, PO2, PO3
Description: This course focused on the design and applications of integrated circuits (ICs), covering topics like operational amplifiers, digital ICs, and their use in real-world electronic systems.
Impact: Gained a deeper understanding of IC design and its applications in various devices, which is fundamental for modern electronics and communication systems.
Skills Developed: IC design, analog and digital circuit analysis, and the application of ICs in system-level electronics.

Course Title: Microprocessors and Microcontrollers
SDG(s): SDG 9
Program Outcomes: PO1, PO2, PO5
Description: This course covered the architecture, programming, and applications of microprocessors and microcontrollers, with an emphasis on embedded systems and interfacing techniques.
Impact: Developed hands-on experience with microcontroller-based projects, essential for designing embedded systems and automating processes in modern electronic applications.
Skills Developed: Embedded systems programming, microprocessor architecture, interfacing with external peripherals, and system-level design.

Course Title: Digital Communications
SDG(s): SDG 9
Program Outcomes: PO1, PO2, PO5
Description: This course focused on the principles of digital communication systems, covering topics such as digital modulation techniques, error control, and signal processing in communication systems.
Impact: Gained a strong understanding of digital communication methods, which are fundamental for the design and analysis of modern communication systems like cellular networks and Wi-Fi.
Skills Developed: Digital modulation, signal processing, error correction techniques, and performance analysis of communication systems.

Course Title: Antennas and Wave Propagation
SDG(s): SDG 9
Program Outcomes: PO1, PO2, PO5
Description: This course focused on the theory and design of antennas, wave propagation, and the factors affecting the performance of communication systems over different media.
Impact: Developed an understanding of how antennas function and how waves propagate, which is crucial for designing efficient wireless communication systems.
Skills Developed: Antenna design, wave propagation analysis, understanding of wireless communication channels, and radio frequency (RF) principles.

Course Title: Professional Communication Skills
SDG(s): SDG 4
Program Outcomes: PO10
Description: This course focused on enhancing communication skills, including writing, speaking, and listening, tailored for professional and technical environments.
Impact: Improved ability to effectively communicate complex ideas in both written and oral forms, which is essential for teamwork and professional presentations.
Skills Developed: Technical writing, public speaking, presentation skills, and effective interpersonal communication.

Course Title: Indian Constitution
SDG(s): SDG 16
Program Outcomes: PO6
Description: This course covered the key principles of the Indian Constitution, including fundamental rights, duties, governance structures, and the role of law in society.
Impact: Increased understanding of legal and constitutional frameworks, fostering responsible citizenship and ethical behavior in professional and societal settings.
Skills Developed: Legal awareness, understanding of governance systems, and the ability to apply constitutional principles in various contexts.

Course Title: CMOS VLSI Design
SDG(s): SDG 9
Program Outcomes: PO1, PO2, PO5
Description: This course focused on the design and fabrication of CMOS (Complementary Metal-Oxide-Semiconductor) circuits and their applications in Very Large Scale Integration (VLSI) systems.
Impact: Gained knowledge in designing power-efficient and high-performance integrated circuits, essential for modern electronic devices such as processors, memory systems, and communication hardware.
Skills Developed: CMOS logic design, VLSI circuit layout, design optimization for power, speed, and area, and understanding of semiconductor technology.

Course Title: Digital Signal Processing
SDG(s): SDG 9
Program Outcomes: PO1, PO2, PO5
Description: This course covered the fundamental techniques in digital signal processing (DSP), including signal representation, filtering, Fourier transforms, and applications in communication and audio systems.
Impact: Developed the ability to design and implement DSP algorithms for various applications, improving the efficiency and performance of digital systems.
Skills Developed: Signal processing techniques, Fourier analysis, filter design, algorithm implementation, and application of DSP in real-world systems.

Course Title: Microwave Engineering
SDG(s): SDG 9
Program Outcomes: PO1, PO2, PO5
Description: This course focused on the principles of microwave transmission, components, and systems, including waveguides, antennas, and microwave circuits used in communication and radar systems.
Impact: Provided a deep understanding of microwave technology, which is essential for the design of high-frequency communication systems, radar systems, and satellite communications.
Skills Developed: Microwave transmission, design of microwave circuits, waveguide analysis, and the application of microwave components in communication systems.

Course Title: Essence of Indian Traditional Knowledge
SDG(s): SDG 4
Program Outcomes: PO6
Description: This course explored the rich heritage of Indian traditional knowledge systems in fields like philosophy, arts, science, and technology, emphasizing their relevance to modern-day problem-solving.
Impact: Fostered an understanding of the value of traditional knowledge and its application in contemporary engineering and societal contexts, promoting ethical and sustainable development.
Skills Developed: Cultural awareness, ethical reasoning, and the ability to integrate traditional wisdom with modern technological advancements.

Course Title: Real Time Embedded Systems
SDG(s): SDG 9
Program Outcomes: PO1, PO2, PO5
Description: This course focused on the design and development of embedded systems with real-time constraints, covering microcontrollers, real-time operating systems, and interfacing techniques.
Impact: Strengthened the ability to build efficient and reliable embedded systems for applications in automation, robotics, and IoT.
Skills Developed: Embedded C programming, real-time task scheduling, hardware-software integration, and microcontroller interfacing.

Course Title: Cellular and Mobile Communications
SDG(s): SDG 9
Program Outcomes: PO1, PO2, PO5
Description: This course covered the principles of cellular systems, mobile radio propagation, multiple access techniques, and modern wireless communication standards.
Impact: Provided a comprehensive understanding of mobile communication systems, essential for developing wireless technologies and improving connectivity.
Skills Developed: Cellular network planning, mobile communication protocols, handoff strategies, and wireless system design.

Course Title: System Verilog for Verification
SDG(s): SDG 9
Program Outcomes: PO2, PO4, PO5
Description: This course introduced advanced verification methodologies using SystemVerilog, focusing on testbench architecture, assertions, and object-oriented programming.
Impact: Enabled the creation of reusable, scalable verification environments and improved functional coverage in hardware design.
Skills Developed: Testbench development, OOP for verification, functional coverage, assertions, and constrained random stimulus generation.

Course Title: Digital Image Processing
SDG(s): SDG 9
Program Outcomes: PO1, PO2, PO5
Description: This course focused on image enhancement, restoration, compression, segmentation, and representation techniques for digital images, along with practical applications in various domains.
Impact: Enhanced understanding of image analysis and processing, enabling the development of efficient visual systems in engineering and technology.
Skills Developed: Image filtering, histogram processing, edge detection, morphological operations, and image compression techniques.

3.2 Case Studies

Case Study Title: Customer Segmentation using K-Means Clustering for E-commerce
SDG(s): SDG 9, SDG 12
Program Outcomes: PO2, PO4, PO5
Description: Case study on applying K-Means clustering to segment e-commerce customers based on purchasing behavior for targeted marketing strategies.
Impact: Enabled data-driven decision-making and improved customer engagement through personalized experiences.
Skills Developed: Data preprocessing, clustering algorithms, customer behavior analysis, and model evaluation.

3.3 Workshops and Industrial Visits

Industrial Visit Title: Visit to BSNL Hyderabad
SDG(s) Addressed: SDG 9 (Industry, Innovation, and Infrastructure)
Description: Visited BSNL Hyderabad to gain insights into the telecommunication infrastructure and services provided by the government-owned telecom company.
Impact: Gained a deeper understanding of the working of telecommunications networks and their role in modern communication systems.
Skills Developed: Telecom network design, understanding of communication protocols, and hands-on experience with telecom infrastructure.

3.4 Mini Project

Project Title: Soil Texture Classification Using ResNet-152
SDG(s) Addressed: SDG 2 (Zero Hunger), SDG 9 (Industry, Innovation, and Infrastructure)
Program Outcomes Mapped: PO2, PO3, PO4, PO5
Description: This publication presents a research study on classifying soil textures using the deep learning model ResNet-152. The work involved creating a comprehensive dataset of soil images and training a deep convolutional neural network to accurately categorize different soil types based on texture and visual characteristics. The objective was to enhance agricultural efficiency through intelligent soil analysis, supporting precision farming practices.
Impact: The study demonstrated the application of deep learning in agriculture, promoting smarter farming techniques through automated soil assessment. It contributes to improved crop planning and sustainable land use by providing a scalable solution for soil classification using image-based data.
Skills Developed: Deep learning, computer vision, model training and evaluation, dataset preparation, agricultural technology, and technical research.

3.5 Cultural and Sports Activities

Activity Title: Volunteering Support for Cultural Events at Campus.
Date/Duration: Ongoing throughout academic tenure (2021–2025).
SDG(s) Addressed: SDG 3 (Good Health and Well-being), SDG 4 (Quality Education), SDG 11 (Sustainable Cities and Communities).
Description: Contributed as a volunteer in various cultural activities organized by the institution.
Impact: Helped create an inclusive and well-organized environment for participants and attendees. Encouraged the smooth execution of events that promoted student creativity, physical well-being, and team spirit.
Skills Developed: Event coordination, teamwork, time management, interpersonal communication.

3.6 Internships and Trainings

3.6.1 Internship - 1

Internship Title: Completion of 6-Week Online Training on VLSI Design in INTERNSHALA
Date/Duration: 2023-6-02 to 2023-7-08
SDG(s) Addressed: SDG 9 (Industry, Innovation, and Infrastructure), SDG 4 (Quality Education)
Description: Successfully completed a comprehensive 6-week online training program on VLSI Design. The training covered a range of topics such as Digital Design, CMOS Logic, VLSI Design Flow, HDL Programming (Verilog/VHDL), Synthesis and Simulation, Physical Design, and a Final Project. The final assessment consisted of practical exercises and design simulations, with a score of 47% obtained.
Impact: Gained foundational knowledge of VLSI design methodologies, tools, and workflows. The training provided a practical understanding of integrated circuit design and verification, bridging academic concepts with industry-relevant skills essential for careers in semiconductor and chip design.
Skills Developed: Digital logic design, Verilog HDL, synthesis and simulation using EDA tools, timing analysis, physical design principles, and debugging of VLSI circuits.

3.6.2 Internship - 2

Internship Title: Online Internship in Cyber Security and Ethical Hacking at Techno Hacks
SDG(s) Addressed: SDG 9 (Industry, Innovation, and Infrastructure), SDG 16 (Peace, Justice, and Strong Institutions)
Description: This 8-week online internship focused on the fundamentals and hands-on aspects of cyber security and ethical hacking. The program covered key areas such as network security, cryptography, vulnerability assessment, penetration testing, malware analysis, and ethical hacking practices. Participants were introduced to various tools including Wireshark, Nmap, Metasploit, and Burp Suite. Through guided labs and real-world case studies, students explored how to identify, analyze, and mitigate security threats effectively.
Impact: This internship built a foundational understanding of cybersecurity principles and enhanced awareness of digital threats and defense mechanisms. It fostered practical knowledge of ethical hacking, helping learners recognize system vulnerabilities and apply mitigation strategies—skills vital for ensuring data protection and system integrity in the modern digital era.
Skills Developed: Network security, penetration testing, ethical hacking, vulnerability scanning, cryptography, malware analysis, Kali Linux tools, cybersecurity best practices, digital forensics, security risk assessment.

3.7 Value-Added Courses

Value Added Course Title: Python for Data Science.
Date/Duration: January - February 2024 (4 Weeks)
SDG(s) Addressed: SDG 4 (Quality Education), SDG 8 (Decent Work and Economic Growth), SDG 9 (Industry, Innovation, and Infrastructure).

Program Outcomes Mapped: PO1 – Engineering Knowledge, PO2 – Problem Analysis, PO3 – Design/Development of Solutions.
Description: Participated in a hands-on value-added course focused on Python for Data Science. The course covered Python programming fundamentals, data analysis libraries (such as NumPy, Pandas, and Matplotlib), and machine learning algorithms. Emphasis was placed on real-world applications of data science in various industries such as healthcare, finance, and technology.
Impact: Enhanced knowledge of data manipulation and analysis using Python, with a focus on solving real-world problems through data-driven decision-making. Gained practical skills that can be directly applied to industries leveraging big data and machine learning technologies.
Skills Developed: Python programming, data wrangling, data visualization, statistical analysis, machine learning algorithms, data preprocessing, and problem-solving with data science tools.

3.8 Course-End Projects

Project Title: Moisture Sensor in Electronic Circuit Analysis (A7405)
Date/Duration: August – October 2023 (2 Months)
SDG(s) Addressed: SDG 2 (Zero Hunger), SDG 6 (Clean Water and Sanitation), SDG 9 (Industry, Innovation, and Infrastructure), SDG 12 (Responsible Consumption and Production)
Program Outcomes Mapped: PO1 – Engineering Knowledge, PO2 – Problem Analysis, PO3 – Design/Development of Solutions, PO4 – Investigation of Complex Problems
Description: Designed and implemented a soil moisture sensor circuit using analog components such as transistors, resistors, and operational amplifiers. The sensor measured soil resistance variations due to moisture levels and converted them into voltage levels to indicate dryness or wetness. The circuit could be interfaced with an indicator or irrigation controller, promoting automation in agriculture.
Impact: Provided practical experience in circuit analysis and sensor interfacing, while promoting sustainable agricultural practices through optimized water usage. Highlighted the use of electronics in solving real-world environmental challenges.
Skills Developed: Analog circuit design, sensor interfacing, transistor switching, op-amp usage, prototyping, and applying theoretical electronics concepts to practical systems.

Project Title: SAVITZKY - GOLAY SMOOTHING FILTER in DSP(A7420).
Date/Duration: January – March 2024 (3 Months)
SDG(s) Addressed: SDG 3 (Good Health and Well-being), SDG 4 (Quality Education), SDG 8 (Decent Work and Economic Growth), SDG 9 (Industry, Innovation, and Infrastructure), SDG 11 (Sustainable Cities and Communities).
Program Outcomes Mapped: PO1 – Engineering Knowledge, PO2 – Problem Analysis, PO3 – Design/Development of Solutions, PO5 – Modern Tool Usage.
Description: Designed and implemented a Savitzky-Golay smoothing filter in MATLAB as part of the DSP course project. The filter was applied to noisy datasets to enhance signal clarity while preserving essential features such as peak sharpness. Explored various polynomial degrees and window lengths to analyze their effect on filtering performance in real-time signal applications.

Impact: Improved understanding of digital signal processing techniques with practical application to healthcare (e.g., ECG signal filtering), industrial monitoring, and smart city systems. Enabled better decision-making in embedded systems through reliable and noise-free signal analysis.
Skills Developed: Digital signal processing fundamentals, MATLAB programming, filter design and analysis, data visualization, real-time signal evaluation, and application of theoretical DSP concepts in practical scenarios.

3.9 Social Service and Community Projects

Project Title: TRUE RADIATIONS – Educational and Career Guidance Platform
Date/Duration: January – April 2024 (4 Months)
SDG(s) Addressed: SDG 4 (Quality Education), SDG 8 (Decent Work and Economic Growth), SDG 10 (Reduced Inequalities)
Program Outcomes Mapped: PO1 – Engineering Knowledge, PO3 – Design/Development of Solutions, PO6 – The Engineer and Society, PO7 – Environment and Sustainability
Description: Developed an interactive website platform named *True Radiations* aimed at supporting students in discovering their educational and psychological potential. The platform serves as a digital guide, offering career counseling, college information, and academic resources. It assists users in making informed decisions regarding course selection and college admission. The platform also emphasizes student empowerment through access to relevant and structured guidance content.
Impact: Promoted informed decision-making among students by providing a centralized platform for career and educational guidance. Bridged the information gap between students and educational institutions, contributing to student success and societal value through better academic alignment.
Skills Developed: Web development, user experience design, educational content structuring, problem-solving, data organization, and community-oriented platform development.

3.10 Major Project

Project Title: Deep Learning Approach for Brain Tumor Classification
Date/Duration: January – April 2025 (4 Months)
SDG(s) Addressed: SDG 3 (Good Health and Well-being), SDG 4 (Quality Education), SDG 9 (Industry, Innovation, and Infrastructure)
Program Outcomes Mapped: PO1 – Engineering Knowledge, PO2 – Problem Analysis, PO3 – Design/Development of Solutions, PO5 – Modern Tool Usage, PO7 – Environment and Sustainability
Description: Developed a deep learning-based model for classifying brain tumors from medical imaging data. Leveraged convolutional neural networks (CNNs) to extract spatial features from MRI scans and accurately differentiate between tumor types. The model was trained and tested using labeled datasets to evaluate performance metrics such as accuracy, precision, and recall. The project aimed to assist healthcare professionals in early and reliable diagnosis, thereby improving treatment outcomes.
Impact: Contributed to advancements in medical imaging and diagnostic technology

by providing an automated, efficient, and accurate tool for brain tumor classification. Helped reduce human error and improve the speed of diagnosis in clinical settings, supporting improved patient care and treatment planning.
Skills Developed: Deep learning, CNNs, medical image processing, Python programming, use of frameworks such as TensorFlow/Keras, model training and evaluation, dataset preprocessing, healthcare AI applications.

3.11 Research Work and Publications

Published Mini Project at 2025 6th International Conference for Emerging Technology (INCET) Project Title: Soil Texture Classification Using ResNet-152
SDG(s) Addressed: SDG 2 (Zero Hunger), SDG 9 (Industry, Innovation, and Infrastructure)
Program Outcomes Mapped: PO2, PO3, PO4, PO5
Description: This publication presents a research study on classifying soil textures using the deep learning model ResNet-152. The work involved creating a comprehensive dataset of soil images and training a deep convolutional neural network to accurately categorize different soil types based on texture and visual characteristics. The objective was to enhance agricultural efficiency through intelligent soil analysis, supporting precision farming practices.
Impact: The study demonstrated the application of deep learning in agriculture, promoting smarter farming techniques through automated soil assessment. It contributes to improved crop planning and sustainable land use by providing a scalable solution for soil classification using image-based data.
Skills Developed: Deep learning, computer vision, model training and evaluation, dataset preparation, agricultural technology, and technical research.

4 Reflection on Learning Experiences

Reflecting on my learning journey, I have come to appreciate the value of both academic and experiential learning in shaping my personal and professional growth. Each activity, whether it was a classroom assignment, a team project, an internship, or community engagement, offered unique insights and learning opportunities that extended far beyond textbooks.

One of the most significant takeaways has been the ability to connect theoretical knowledge with practical applications. Working on real-world problems during projects and internships allowed me to better understand the relevance of core concepts, and encouraged me to think critically and creatively in finding solutions. This process has strengthened my problem-solving skills and deepened my technical knowledge. Collaborative tasks, such as participating in hackathons and group projects, helped me enhance my communication and teamwork abilities. These experiences taught me how to listen actively, contribute meaningfully, and manage differences constructively skills that are essential in any professional environment.

My involvement in social outreach activities and sustainable development initiatives has been particularly fulfilling. These experiences nurtured a sense of ethical responsibility and reminded me of the impact we, as students and future professionals, can

have on society. They also helped me understand the importance of aligning our learning with global challenges, especially the United Nations Sustainable Development Goals (SDGs). Throughout this journey, I also recognized the importance of self-motivation and continuous learning. Whether it was acquiring new digital tools, researching emerging technologies, or reflecting on feedback, I have developed the habit of lifelong learning a mindset that will continue to guide me as I advance in my career.

In summary, this portfolio reflects not only what I have learned but how I have grown intellectually, socially, and ethically. Each experience has contributed to making me a more informed, capable, and responsible individual.

5 Future Goals

After graduation, I aspire to work in the software industry, particularly in areas that support green technologies and sustainable solutions. I aim to specialize in developing software systems for renewable energy management, smart grids, and eco-efficient infrastructure. Pursuing a master's degree in Sustainable Systems Engineering will help me integrate software innovation with environmental goals. In the long term, I hope to contribute to environmental policy-making through data-driven insights and sustainable software practices. I also intend to mentor aspiring tech students and engage in community initiatives that promote digital solutions aligned with the UN Sustainable Development Goals (SDGs).

Annexure

List of Evidences - Certificates, Appraisals and Reports

#	Description
1	VLSI Design Internship CERTIFICATE
2	Cybersecurity & Ethical Hacking Internship CERTIFICATE



Figure 1: VLSI Design Internship CERTIFICATE.



Figure 2: Cybersecurity and Ethical Hacking Internship CERTIFICATE.